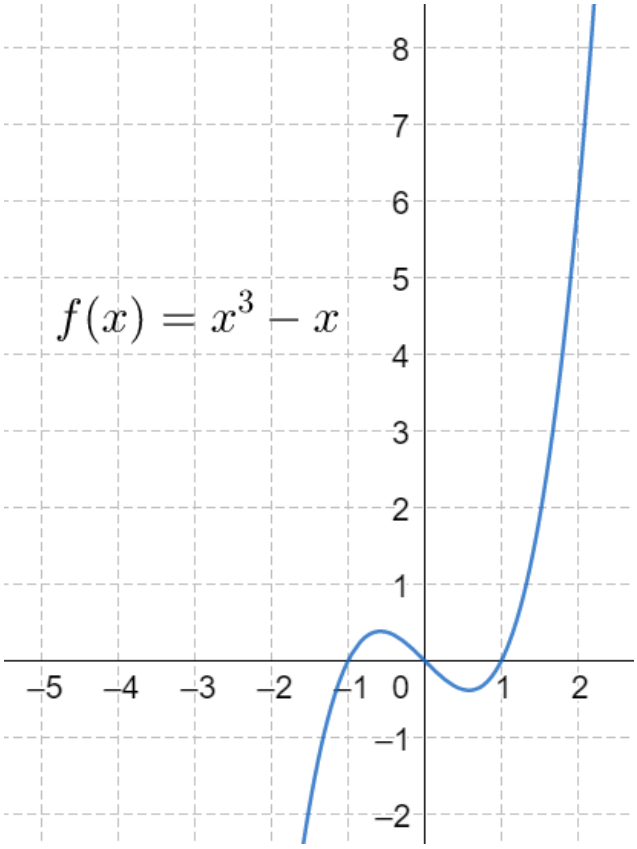
**Rajinder Lal**
**Math 400**

## Section 2.5 Continuity

We will start off with an informal definition of continuity: The graph of a continuous function can be drawn without lifting up your pen. So there are no holes or jumps in the graph.

**Here is an example of the graph of a continuous function:**

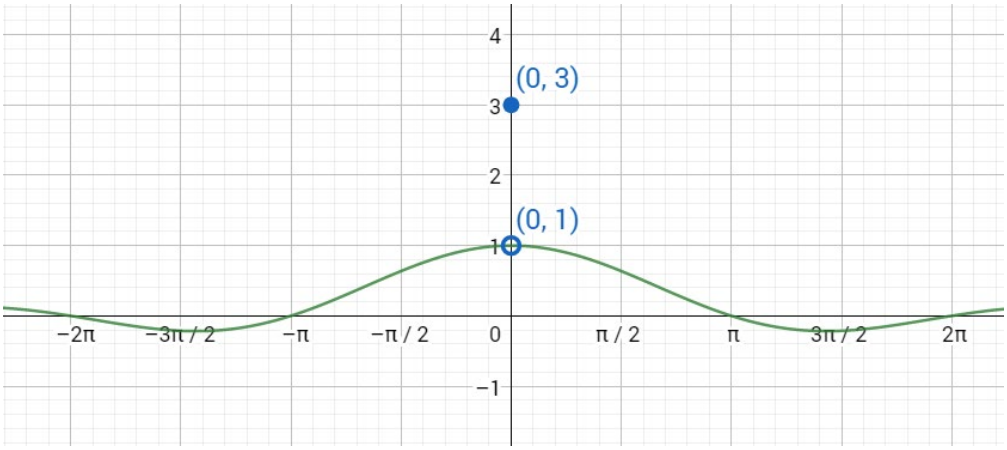


**Objective 1: Types of discontinuities.**

Let's look at examples of graphs of functions that have discontinuities:

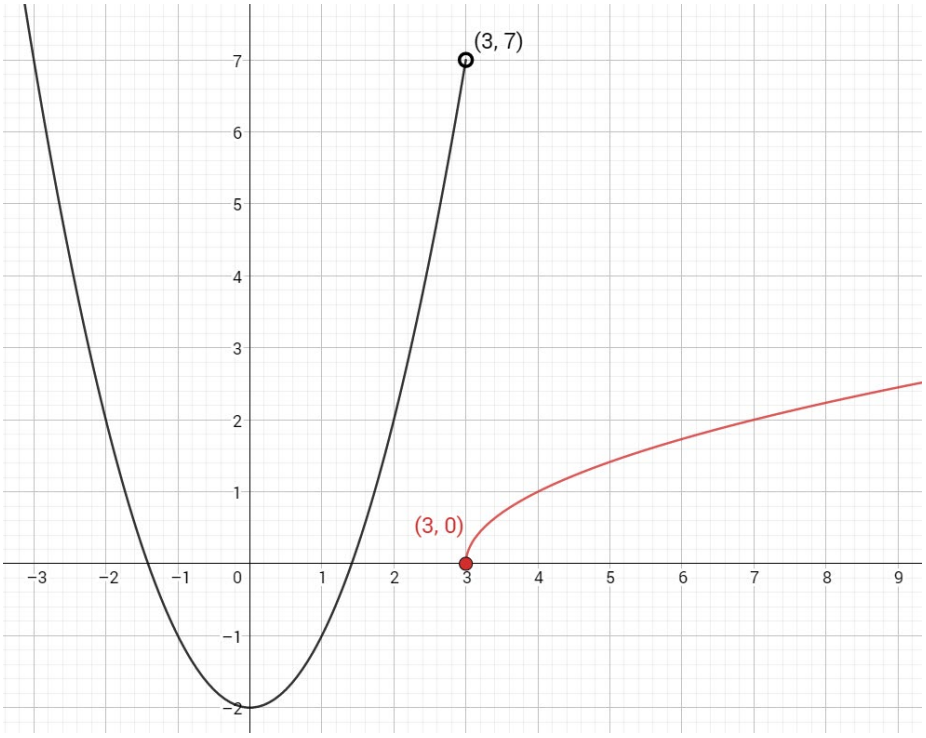
**Example 1:** Removable Discontinuity.

$$f(x)=\begin{cases}\dfrac{\sin x}{x} & \text{, if } x\neq 0\\ 3 & \text{, if } x=0\end{cases}$$



**Example 2:** Jump discontinuity

$$f(x)=\begin{cases}\sqrt{x-3}\ , \text{ if } x \geq 3 \\ x^2-2, \text{ if } x<3\end{cases}$$



**Objective 2: Determine if a function is continous at a number.**

**1 Definition** A function $f$ is **continuous at a number** $a$ if

$$\lim_{x \to a} f(x) = f(a)$$

Notice that Definition 1 implicitly requires three things if *f* is continuous at *a*:

1.

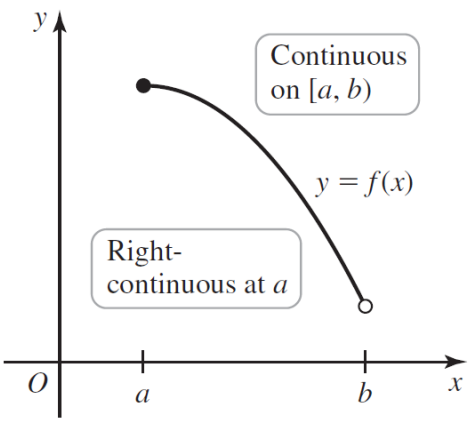
2.

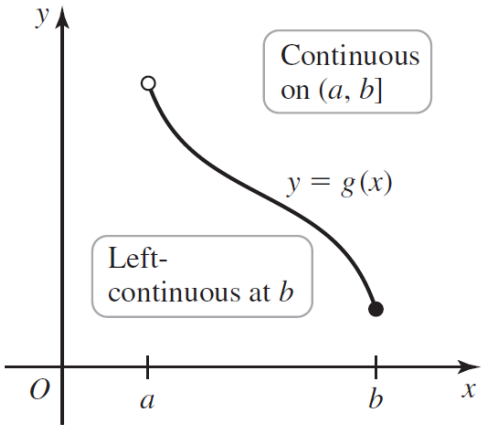
3.

**Example 3:** Determine whether the following function is continuous at a.

$$f(x)=\begin{cases}-2x+1\text{, if } x<0\\ \sqrt{x}+1\text{, if } x\geq 0\end{cases},\quad a=0$$

**Objective 3: Continuity on an interval**




**Definition:** A function $f$ is **continuous from the right at a number** $a$ if

**Definition:** A function $f$ is **continuous from the left at** $b$ if

**3 Definition** A function $f$ is **continuous on an interval** if it is continuous at every number in the interval. (If $f$ is defined only on one side of an endpoint of the interval, we understand *continuous* at the endpoint to mean *continuous from the right* or *continuous from the left*.)

**7 Theorem** The following types of functions are continuous at every number in their domains:

- polynomials
- rational functions
- root functions
- trigonometric functions
- inverse trigonometric functions
- exponential functions
- logarithmic functions

**4 Theorem** If $f$ and $g$ are continuous at $a$ and $c$ is a constant, then the following functions are also continuous at $a$:

1. $f + g$
2. $f - g$
3. $cf$
4. $fg$
5. $\frac{f}{g}$ if $g(a) \neq 0$

**Example 4:** Find the intervals where the function is continuous and state the values where the function is discontinuous.

$$f(x) = \frac{\cot x}{\sqrt{9 - x^2}}$$

**Objective 4: Composition of continuous functions.**

**8 Theorem** If $f$ is continuous at $b$ and $\lim_{x \to a} g(x) = b$, then $\lim_{x \to a} f(g(x)) = f(b)$. In other words,

$$\lim_{x \to a} f(g(x)) = f\left(\lim_{x \to a} g(x)\right)$$

**Example 5:** Evaluate the limit. (Section 2.5 page 121 Example 8)

$$\lim_{x \to 1} \arcsin\left(\frac{1-\sqrt{x}}{1-x}\right)$$

**9 Theorem** If $g$ is continuous at $a$ and $f$ is continuous at $g(a)$, then the composite function $f \circ g$ given by $(f \circ g)(x) = f(g(x))$ is continuous at $a$.

**Example 6:** Find the interval(s) where the function is continuous.

$$h(x) = \ln\left(x^2 - 1\right)$$

**Example 7:** Use continuity to evaluate the limit.

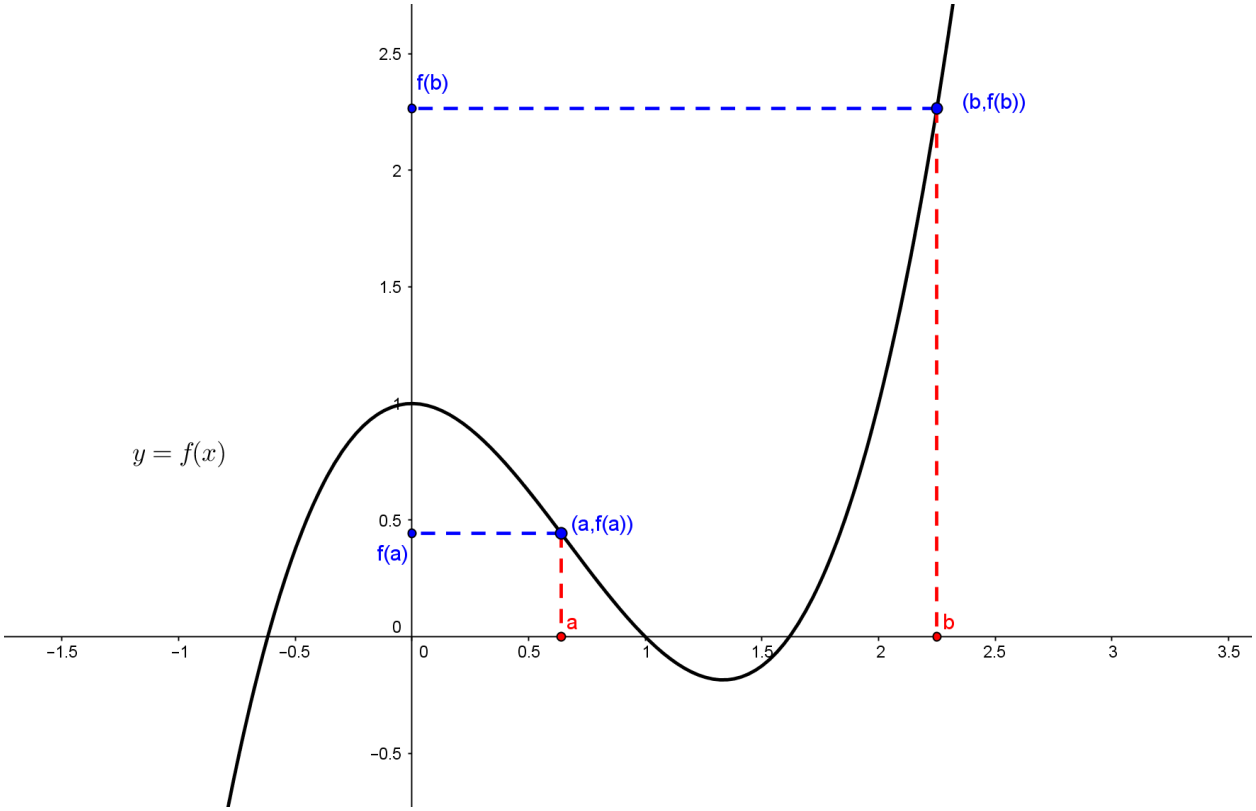
$(a.)\ \lim_{x\to -1} x^2\sqrt{3-x}$

$(b.)\ \lim_{x\to \pi} e^{\sin x}$

$(c.)\ \lim_{x\to 4} \log_7\left(\frac{5x+1}{x-1}\right)$

**Objective 5: Intermediate Value Theorem.**

**10 The Intermediate Value Theorem** Suppose that $f$ is continuous on the closed interval $[a, b]$ and let $N$ be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number $c$ in $(a, b)$ such that $f(c) = N$.



**Example 8:** Use the Intermediate Value Theorem to show that the following equation has a solution on the given interval.

$x \ln x - 1 = 0$ on $(1, e)$